UCLouvain

École polytechnique de Louvain

ÉCOLE POLYTECHNIQUE DE LOUVAIN

LEPL2022 - Health Innovation Classes

Ideation

March 17th, 2026

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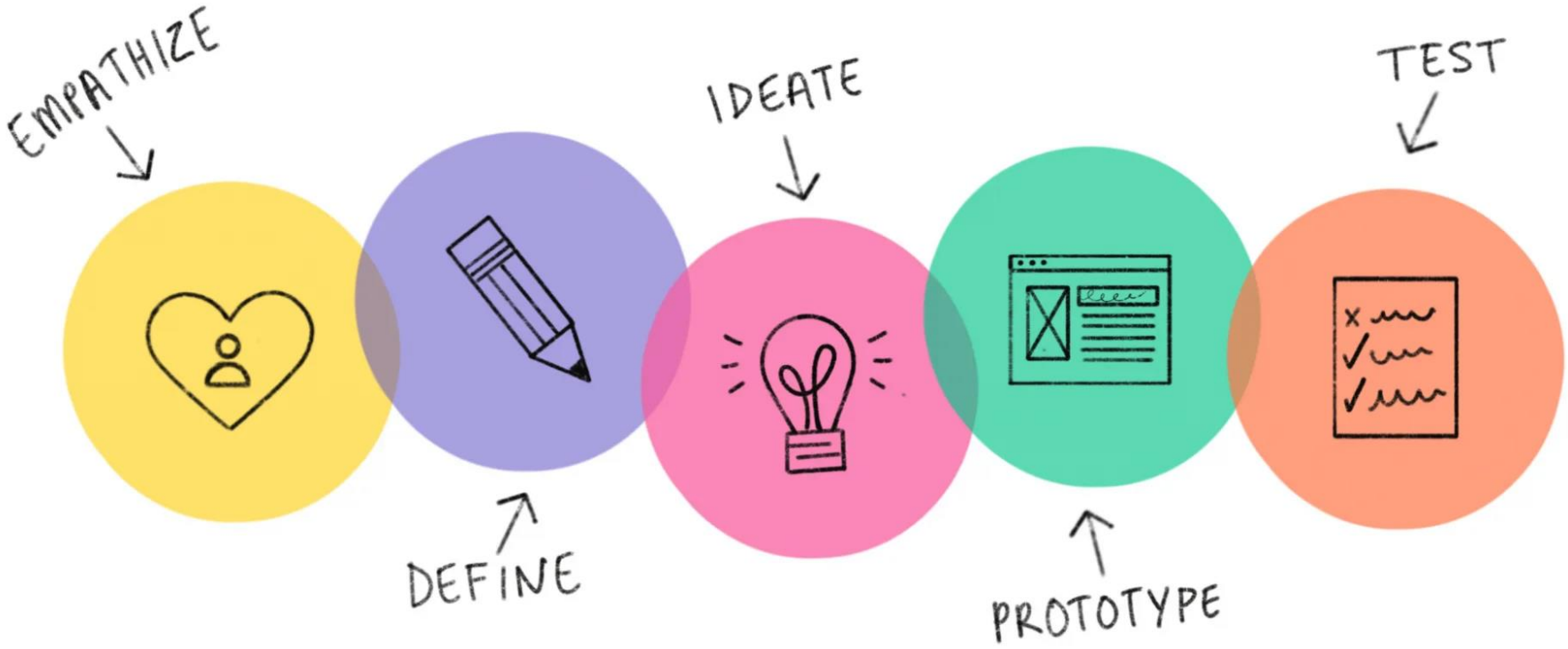
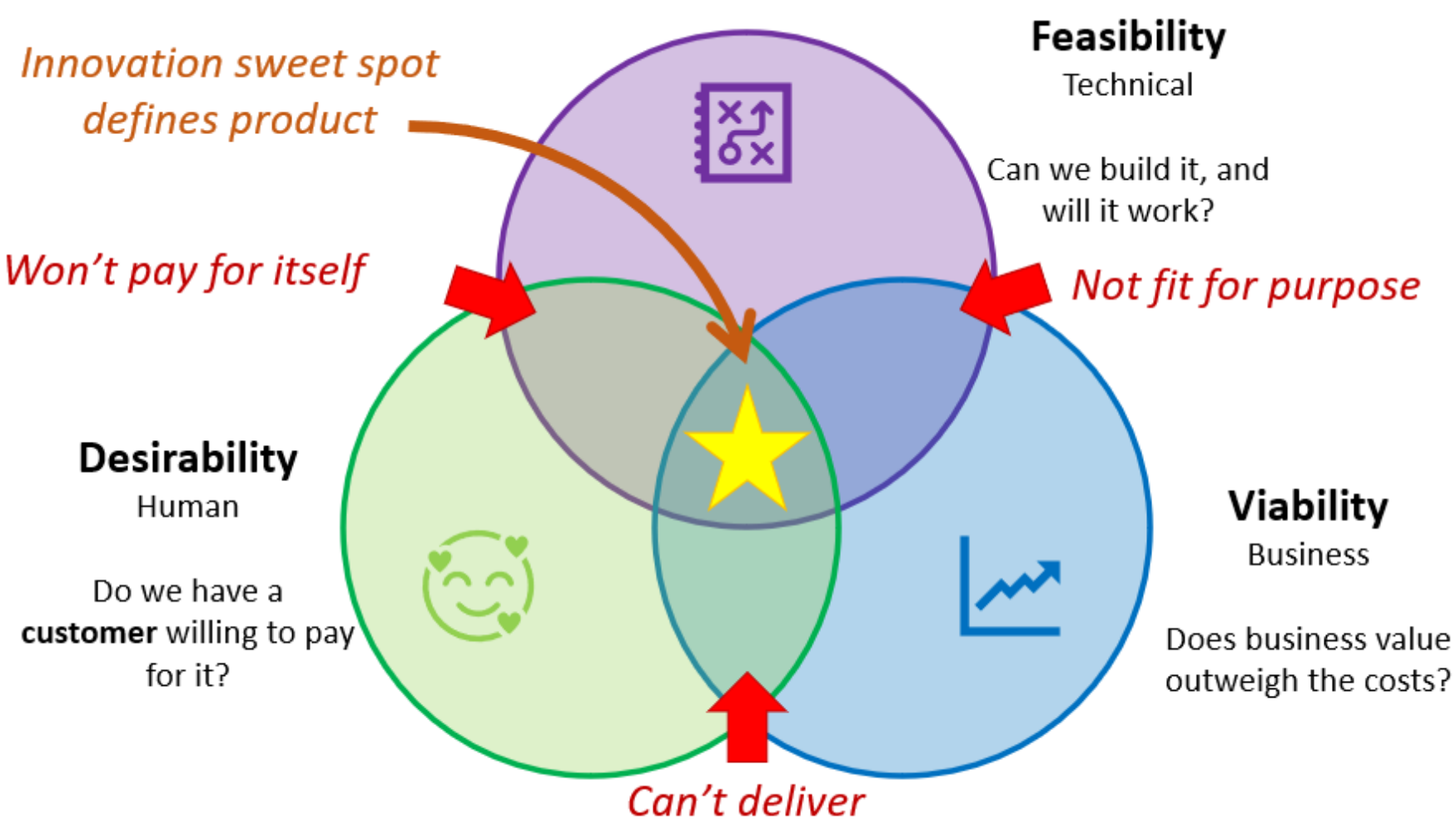
Reminder

DESIGN THINKING PROCESS



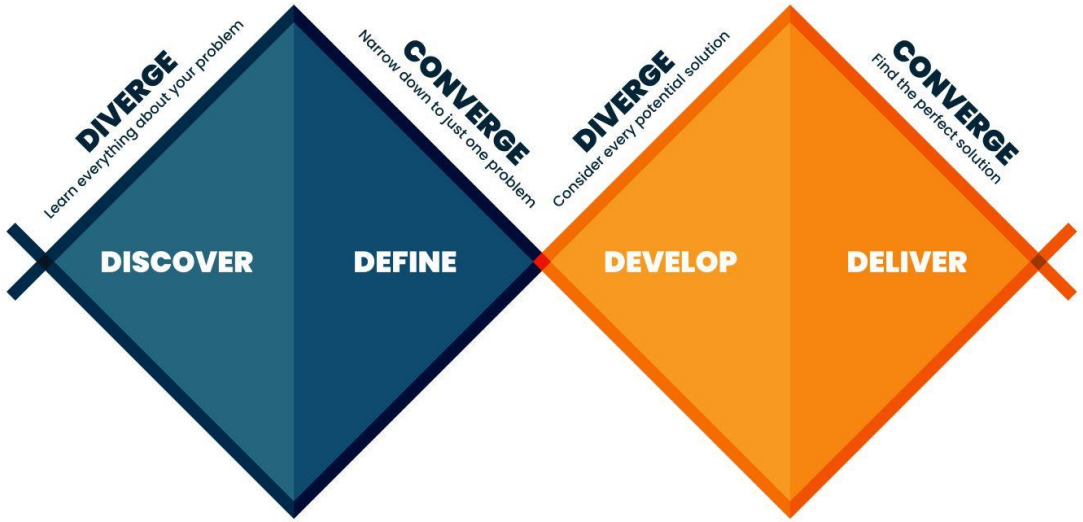

Innovation sweet spot defines product
Feasibility
Technical
Can we build it, and will it work?
Won't pay for itself
Not fit for purpose
Desirability
Human
Do we have a **customer** willing to pay for it?
Viability
Business
Does business value outweigh the costs?
Can't deliver

LEARNING OBJECTIVES FOR TODAY

By the end of the session students should be able to:

- generate many diverse ideas
- use structured ideation tools
- explore solution spaces systematically
- identify promising concepts to prototype

DIVERGENCE AND CONVERGENCE



https://www.fluxspace.io/resources/the-4-ds-double-diamond-design-thinking-model

- Double Diamond
- Diverge, then converge — twice
- First diamond: understand problem
- Second diamond: explore solutions
- Open wide before narrowing
- Avoid premature problem/solution fixation
- Structure uncertainty and learning

IDEATION IS ABOUT FIRST DIVERGING

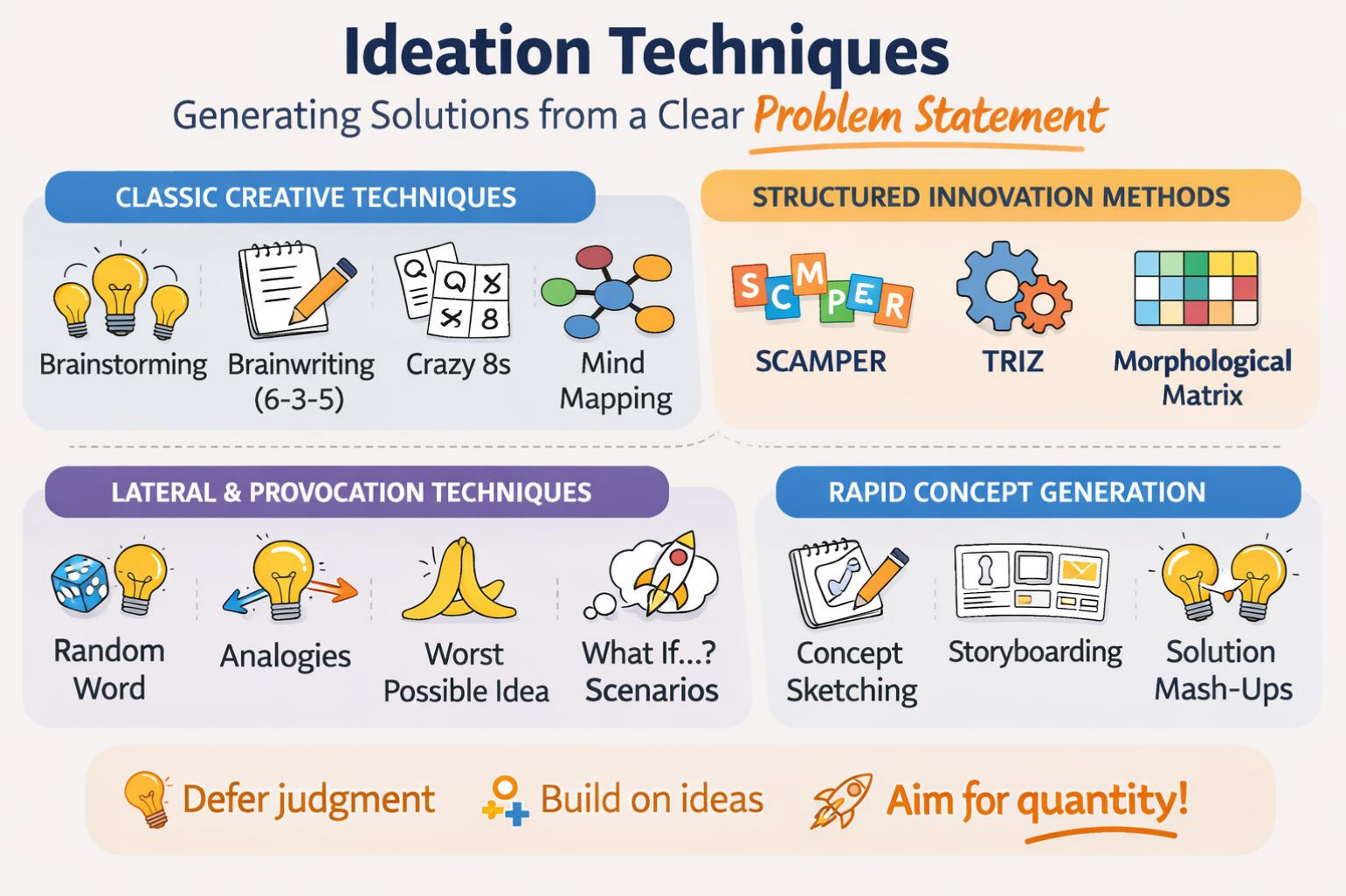
During ideation we intentionally generate **more ideas than we need**.

Quantity leads to **unexpected quality**.

…THEN CONVERGING

Convergence turns creativity into **decisions.**

Many ideas become **one strong solution.**



Ideation techniques work best when combined.

For example, use brainstorm to come with many ideas per SCAMPER dimension then combine using solution mash-ups.

Brainstorming

BRAINSTORMING

- Defer Judgment
- Encourage Wild Ideas
- Build on the Ideas of Others
- Stay Focused on the Topic
- One Conversation at a Time
- Be Visual
- Go for Quantity

SCAMPER

The SCAMPER model

Seven perspectives to provoke creative solutions to challenging problems.

S	C	A	M	P	E	R
Substitute	Combine	Adapt	Modify	Put to another use	Eliminate	Reverse
Replace a part, material, or process with something else.	Join elements, ideas, or functions together in new ways – or find a new element you can merge with.	Modify something to better suit a new purpose, person or context.	Enlarge, reduce, change the shape, or alter attributes. Can a small change have a big effect?	Rather than changing the thing itself, consider changing the context it exists in.	Remove elements, simplify, or pare down to essentials. Is less more?	Flip the script, re-order your priorities, invert cause and effect, and turn it all upside-down.

BiteSize Learning

Challenge	Substitute	Combine	Adapt	Modify	Put to Another Use	Eliminate	Reverse
Invent a better kind of iPhone case	Use eco-friendly materials like bamboo or recycled plastics.	Integrate a wallet or stand feature into the case.	Adapt shock-absorbent materials used in aerospace for better protection.	Modify the case to include a solar panel for charging.	Use the case as a fitness accessory with a built-in pedometer.	Remove unnecessary packaging for a minimalistic, eco-friendly approach.	Design the case to enhance the phone's aesthetics, making it a fashion statement.
Customer queries are taking too long to answer	Replace email responses with a chatbot for common queries.	Combine AI with human oversight for faster, more accurate responses.	Use strategies from fast-food chains to streamline query processing.	Modify the query process with prioritisation algorithms for efficiency.	Use resolved queries as a knowledge base for customer self-service.	Eliminate redundant steps in the query resolution process.	Reverse the process: let customers rate answers and guide the service improvements.
Our tech department has high turnover	Substitute traditional hiring methods with skill-based hiring challenges.	Combine flexible working options with career development programs.	Adapt retention strategies from industries known for loyalty, like hospitality.	Modify job roles to include more cross-training and skill development.	Use exit interviews to create a mentorship program for remaining staff.	Cut down on bureaucratic procedures that may frustrate tech employees.	Reverse the turnover trend by focusing on internal promotions and recognitions.
Alternative to traditional project management software	Use a decentralized, blockchain-based system for project collaboration.	Merge the best features of communication tools with task management software.	Adapt the Kanban method for use in a more visual, intuitive software interface.	Modify the interface to be more user-friendly for non-technical teams.	Repurpose social media platforms for collaborative project management.	Eliminate complex features that are rarely used by most teams.	Switch from a software to a physical board that digital nomads can carry.
Marketing campaign for chocolate cookies	Substitute traditional advertising with user-generated content competitions.	Combine with a popular online game for a themed marketing campaign.	Adapt storytelling techniques from cinema for engaging ads.	Modify the flavour profiles of the cookies in the campaign for uniqueness.	Put the cookies to use in recipes, promoting them as versatile ingredients.	Eliminate the use of artificial ingredients in the cookies.	Reverse the roles: let customers create the campaign, with the best one winning a prize.

EXERCISE

Many elderly patients **forget to take their medication** or take it incorrectly (wrong time, wrong dose), leading to health complications and hospitalizations.

Problem statement:

How might we help elderly patients take their medication safely and regularly at home?

Start from a standard pill box, then explore the 7 alternatives of SCAMPER.

Generate one idea per dimension.

Morphological Analysis

PRINCIPLES

Structured method for exploring all possible solutions of a complex problem by systematically combining alternatives for each function - Developed by Fritz Zwicky

1. Decompose the problem into functions → Break the system into its essential dimensions or functions.

2. List possible solutions for each function → For every function, identify multiple technical options.

3. Build the morphological matrix (morphological box) → Organize the functions and their options in a structured table.

4. Explore combinations systematically → Generate concepts by combining one option from each function.

5. Evaluate and select feasible configurations → Screen combinations to identify innovative and viable solutions.

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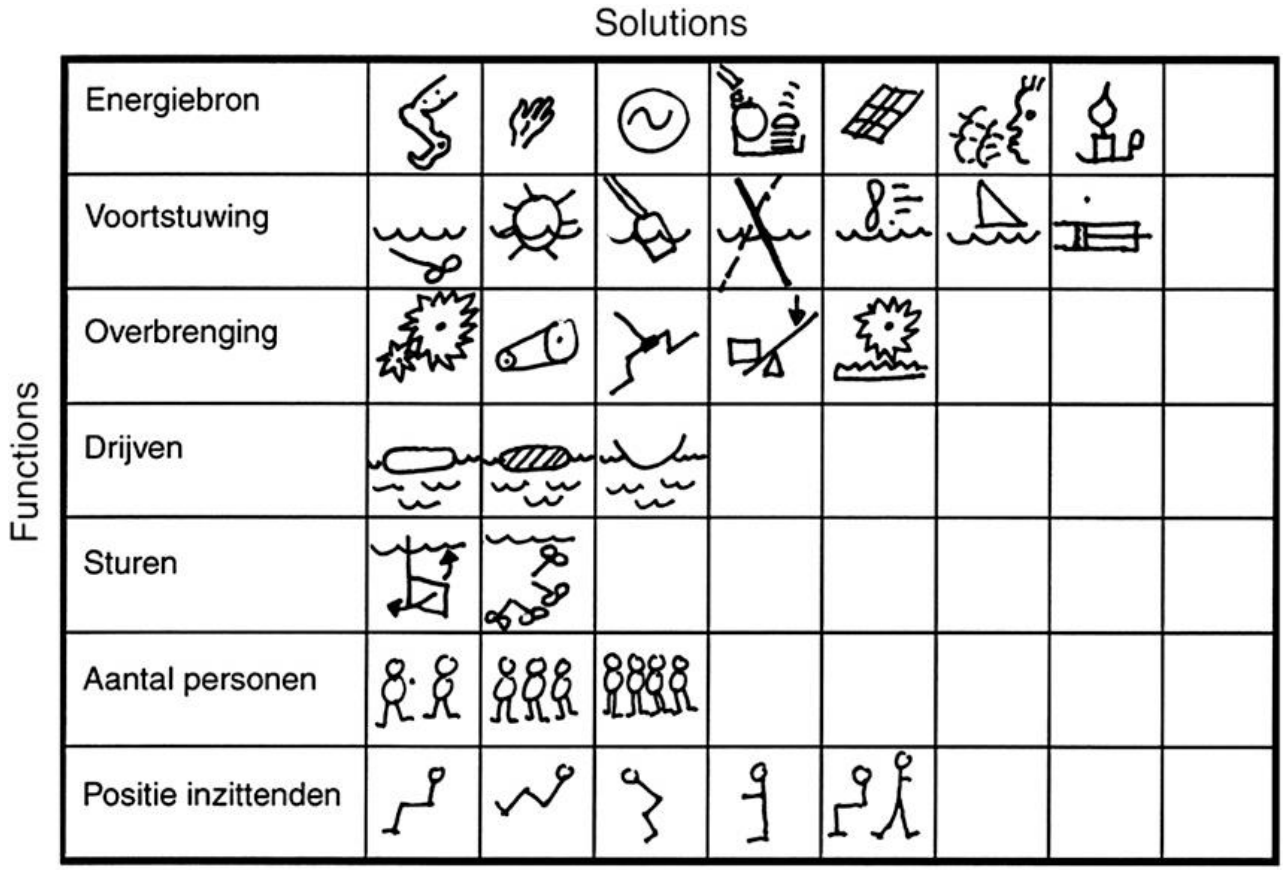
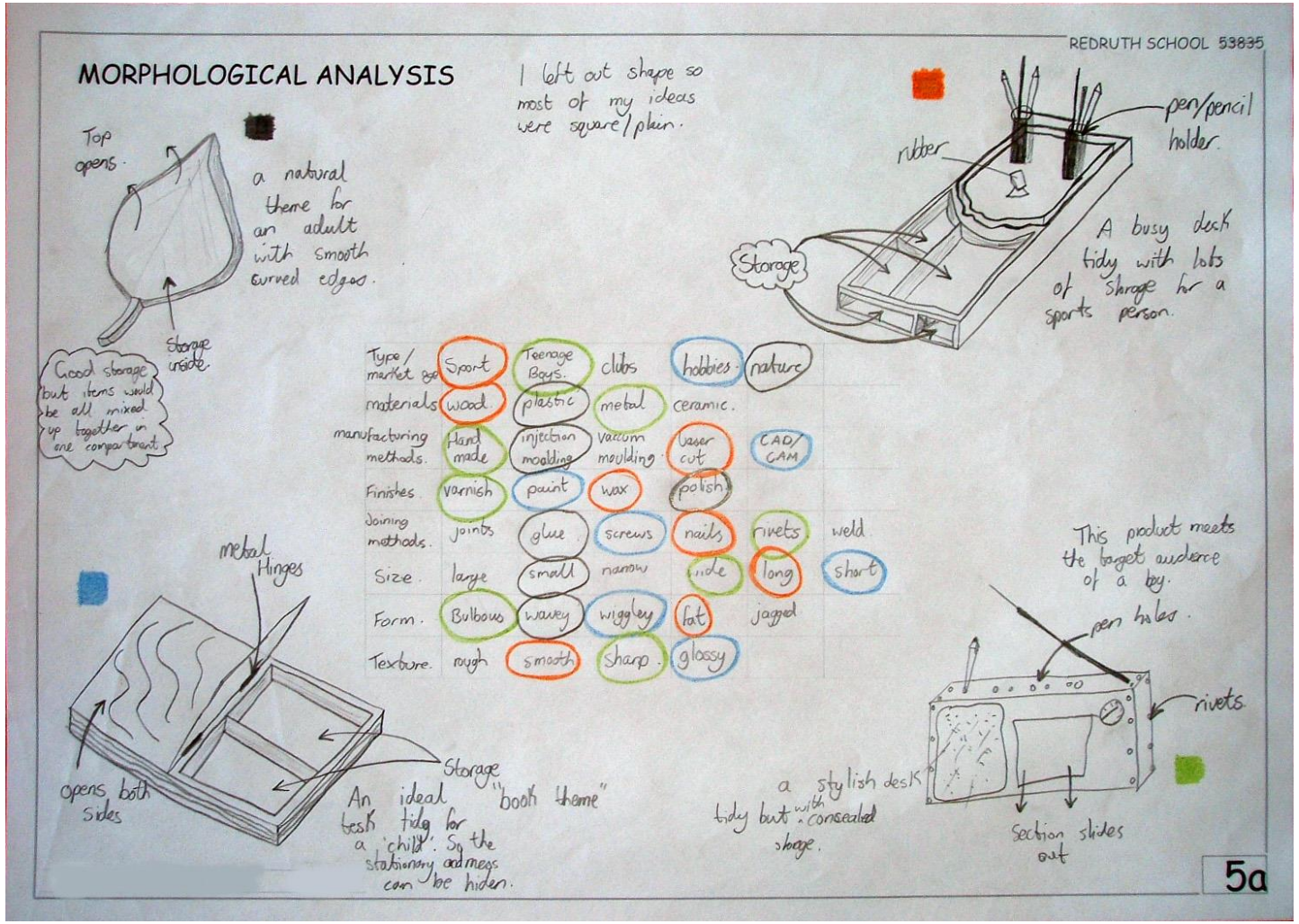
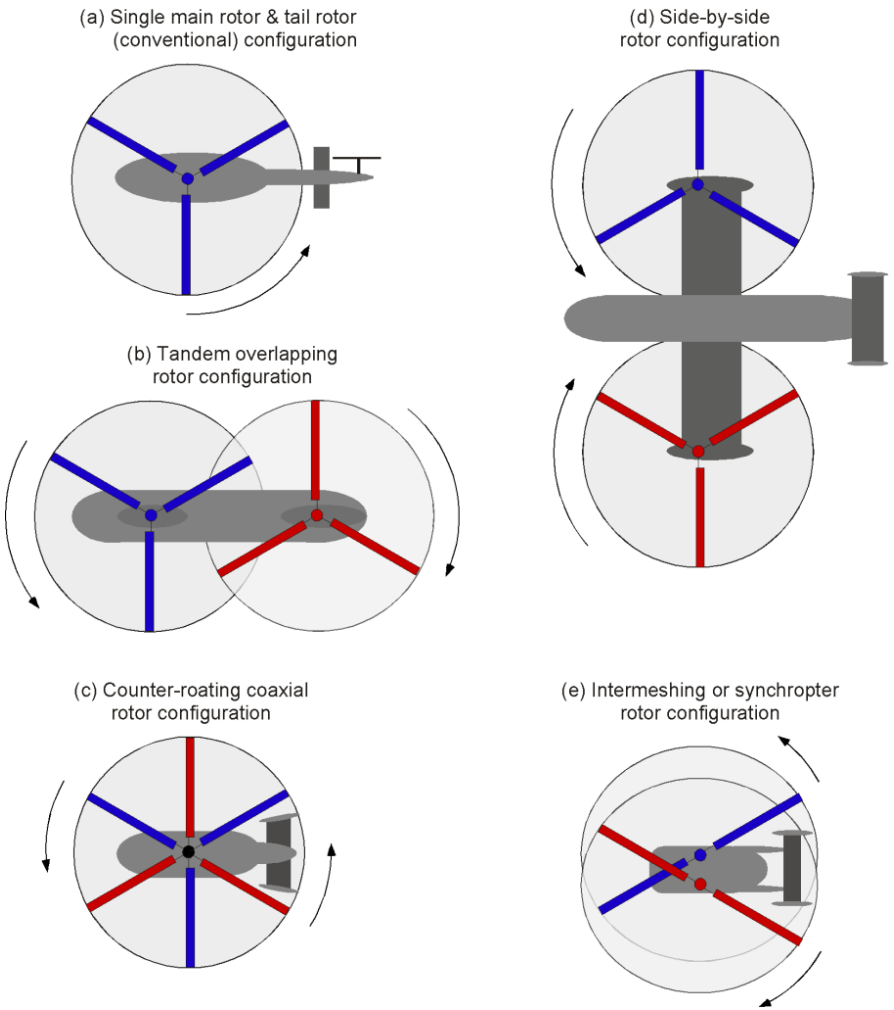


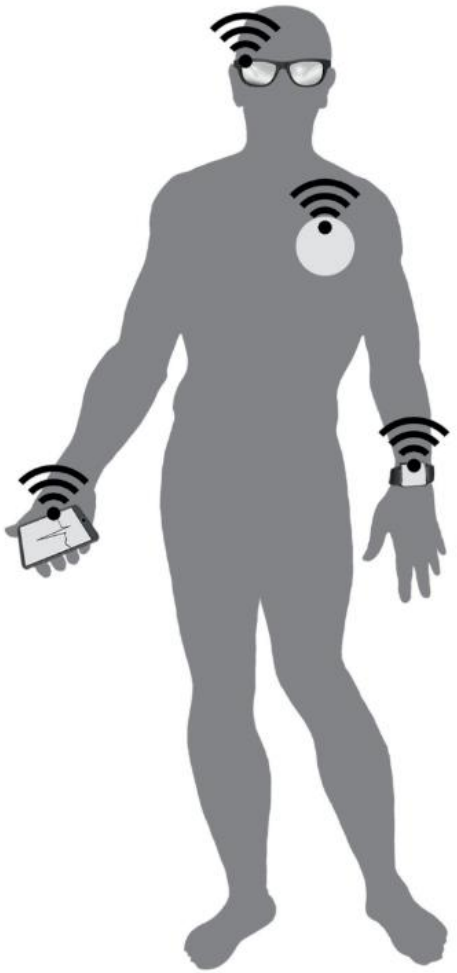

REDRUTH SCHOOL 53835
MORPHOLOGICAL ANALYSIS
I left out shape so most of my ideas were square/plain.
Top opens.
a natural theme for an adult with smooth curved edges.
Storage inside.
Good storage but items would be all mixed up together in one compartment.
pen/pencil holder.
rubber
Storage
A busy desk tidy with lots of storage for a sports person.
Type / market age
Sport
Teenage Boys.
clubs
hobbies
nature
materials
wood.
plastic
metal
ceramic.
manufacturing methods.
Hand made
injection moulding
vacuum moulding
laser cut
CAD/CAM
Finishes.
varnish
paint
wax
polish.
Joining methods.
joints
glue
screws
nails
rivets
weld
Size.
large
small
narrow
wide
long
short
Form.
Bulbous
wavey
wiggley
fat
jagged
Texture.
rough
smooth
sharp.
glossy
metal Hinges
opens both sides
Storage "book theme"
An ideal desk tidy for a 'child'. So the stationary and mags can be hiden.
This product meets the target audience of a boy.
pen holes.
rivets
a stylish desk tidy but with concealed storage.
Section slides out
5a

Function	Possible solutions
Lift generation	single rotor / two rotors / multiple rotors
Torque compensation	tail rotor / coaxial rotors / tandem rotors
Direction control	cyclic pitch / thrust vectoring
Power transmission	central shaft / distributed motors



EXERCISE

Many patients with **chronic diseases (e.g., hypertension or diabetes)** need to **monitor health parameters at home**, but current devices are often **complex, uncomfortable, or poorly integrated into daily life**.



Problem statement:

How might we design a home medical device that makes health monitoring easier and more engaging for patients?

Use Morphological analysis to create 3 different concepts by different combinations of functions. Give each of them a name.

Functions: Body parameter measured - Sensing method - Device form factor - User feedback / interface - Data transmission / sharing

TRIZ

PRINCIPLES

Origin

- Developed by Genrich Altshuller, Soviet engineer and patent examiner
- Started in 1946 after analyzing 200,000+ patents
- Discovered that innovations follow recurring patterns
- Led to the creation of TRIZ (Theory of Inventive Problem Solving)

Fundamental Principle: Contradictions

- Innovation occurs when contradictions are resolved.
- Example: A system must be stronger but also lighter → Improving one parameter degrades another
- TRIZ proposes that true innovation eliminates the contradiction rather than accepting a compromise.

METHOD

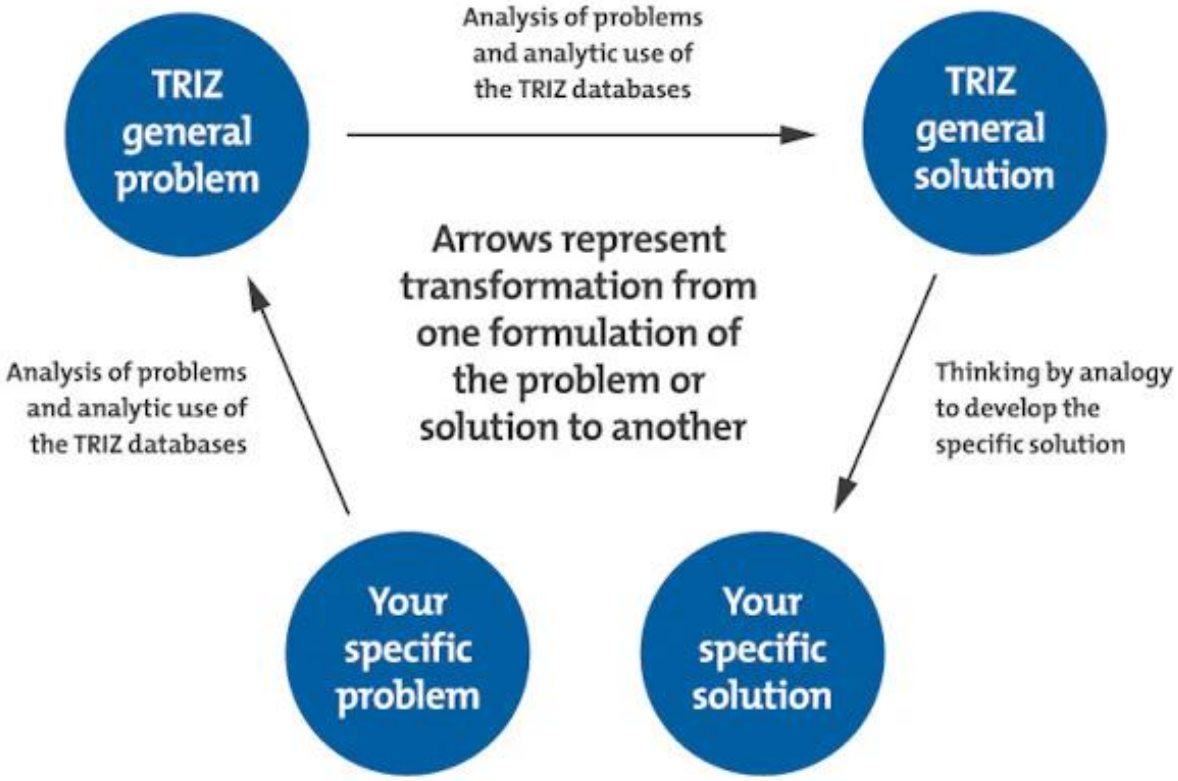

TRIZ general problem
Analysis of problems and analytic use of the TRIZ databases
TRIZ general solution
Arrows represent transformation from one formulation of the problem or solution to another
Analysis of problems and analytic use of the TRIZ databases
Thinking by analogy to develop the specific solution
Your specific problem
Your specific solution

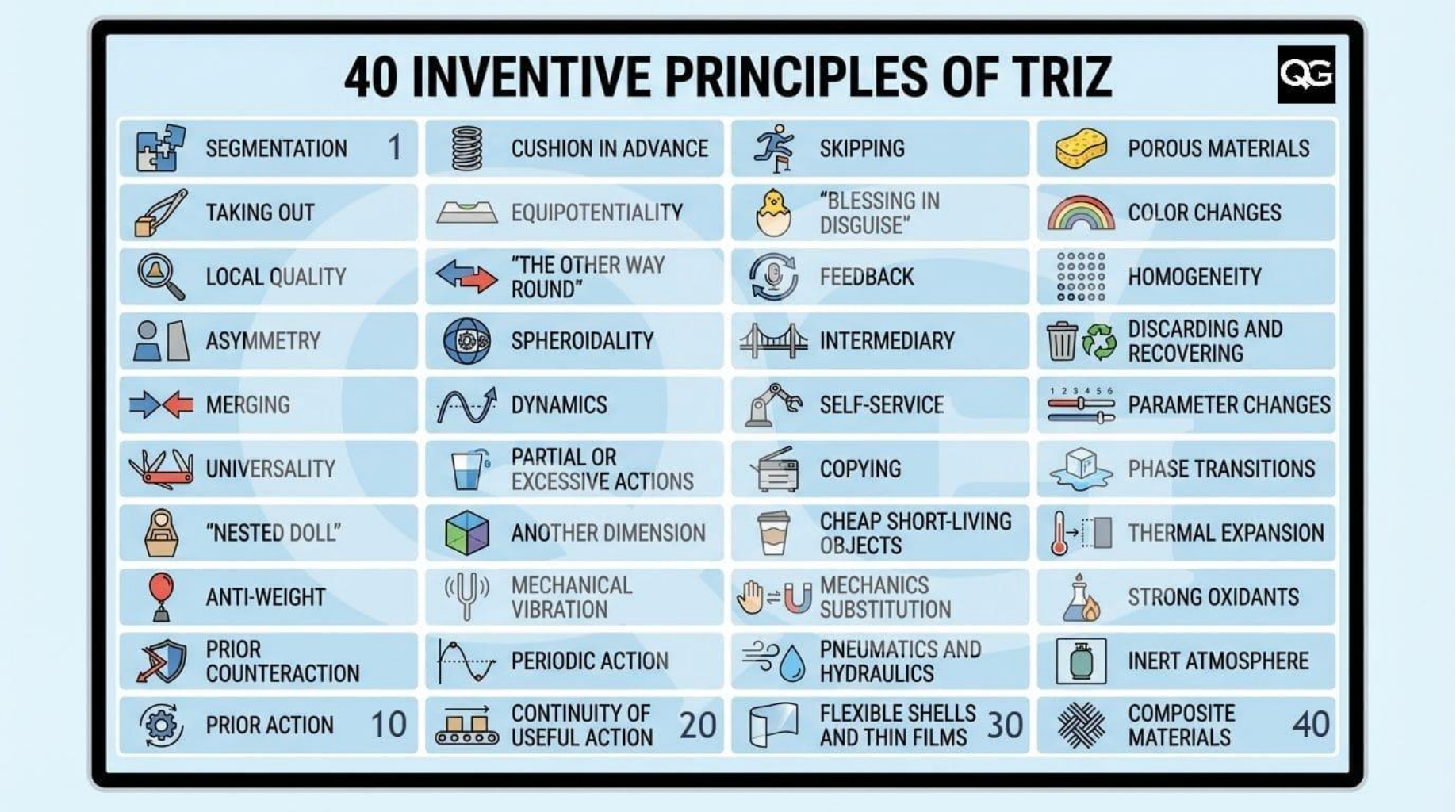
Table 2. Thirty-nine engineering parameters of TRIZ.

No.	Engineering parameter	No.	Engineering parameter
1	Weight of M	21	Power
2	Weight of non-M	22	Waste of energy
3	Length of M	23	Waste of substance
4	Length of non-M	24	Loss of information
5	Area of M	25	Waste of time
6	Area of non-M	26	Amount of substance
7	Volume of M	27	Reliability
8	Volume of non-M	28	Accuracy of measurement
9	Speed	29	Accuracy of manufacture
10	Force	30	Harmful factors acting on object
11	Tension/pressure	31	Harmful side effects
12	Shape	32	Manufacturability
13	Stability of object	33	Convenience of use
14	Strength	34	Reparability
15	Durability of M	35	Adaptability
16	Durability of non-M	36	Complexity of device
17	Temperature	37	Complexity of control
18	Brightness	38	Level of automation
19	Energy spent by M	39	Productivity
20	Energy spent by non-M		

M: moving object.

https://www.triz-consulting.de/wp-content/uploads/2021/09/Contradiction_Matrix_V11.pdf

Worsening Feature → / Improving Feature ↓		Physical								
		Weight of Moving object	Weight of Stationary Object	Length/Angle of Moving Object	Length/Angle of Stationary Object	Area of Moving Object	Area of Stationary Object	Volume of Moving Object	Volume of Stationary Object	Shape
		1	2	3	4	5	6	7	8	9
Physical	1 Weight of Moving Object		3 19 35 40	17 15 8 35	15 17 28 12	28 17 29 35	17 28 1 29	28 29 7 40	40 35 2 4 7	3 35 14 17
	2 Weight of Stationary Object	35 3 40 2		17 4 30 35	17 35 9 31	17 3 30 7	17 14 3 35	14 13 3 40	31 35 7 3	13 7 3 30
	3 Length/Angle of Moving Object	31 4 17 15	1 2 17 15		1 17 15 24	15 17 4 14	17 3 7 15	17 14 7 4 3	17 31 3 19	1 35 29 3
	4 Length/Angle of Stationary Object	35 30 31 8	35 31 40 2	3 1 4 19 17		3 4 19 17	17 40 35 10	35 30 14 7	14 35 17 2	13 14 15 7
	5 Area of Moving Object	31 17 3 4 1	17 15 3 31	14 15 4 18	14 17 15 4		17 1 4 3	14 17 7 4	14 17 7 13	35 4 14 17
	6 Area of Stationary Object	14 31 17 19	35 14 31 30	17 19 3 13	17 14 3 4 7	4 31 7 19		17 18 14 7	14 28 26 13	17 5 4 7


40 INVENTIVE PRINCIPLES OF TRIZ
QG
SEGMENTATION 1
CUSHION IN ADVANCE
SKIPPING
POROUS MATERIALS
TAKING OUT
EQUIPOTENTIALITY
"BLESSING IN DISGUISE"
COLOR CHANGES
LOCAL QUALITY
"THE OTHER WAY ROUND"
FEEDBACK
HOMOGENEITY
ASYMMETRY
SPHEROIDALITY
INTERMEDIARY
DISCARDING AND RECOVERING
MERGING
DYNAMICS
SELF-SERVICE
PARAMETER CHANGES
UNIVERSALITY
PARTIAL OR EXCESSIVE ACTIONS
COPYING
PHASE TRANSITIONS
"NESTED DOLL"
ANOTHER DIMENSION
CHEAP SHORT-LIVING OBJECTS
THERMAL EXPANSION
ANTI-WEIGHT
MECHANICAL VIBRATION
MECHANICS SUBSTITUTION
STRONG OXIDANTS
PRIOR COUNTERACTION
PERIODIC ACTION
PNEUMATICS AND HYDRAULICS
INERT ATMOSPHERE
PRIOR ACTION 10
CONTINUITY OF USEFUL ACTION 20
FLEXIBLE SHELLS AND THIN FILMS 30
COMPOSITE MATERIALS 40

Utility knives become **dull quickly** when cutting cardboard, plastic, or tape.

Two possible solutions existed:

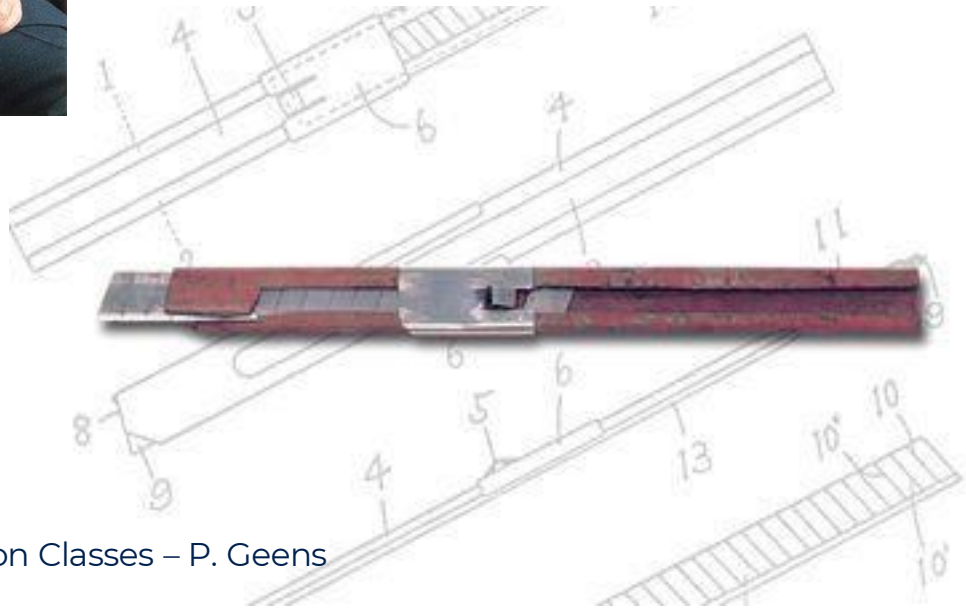
- Replace the blade frequently → **expensive and inconvenient**
- Keep using the same blade → **poor cutting performance**

Maintaining durability

while

Reducing complexity of repair

→ Principle 1: Segmentation



EXERCISE

In hospitals, **doctors spend a lot of time on administrative documentation** (electronic health records, reports, prescriptions).

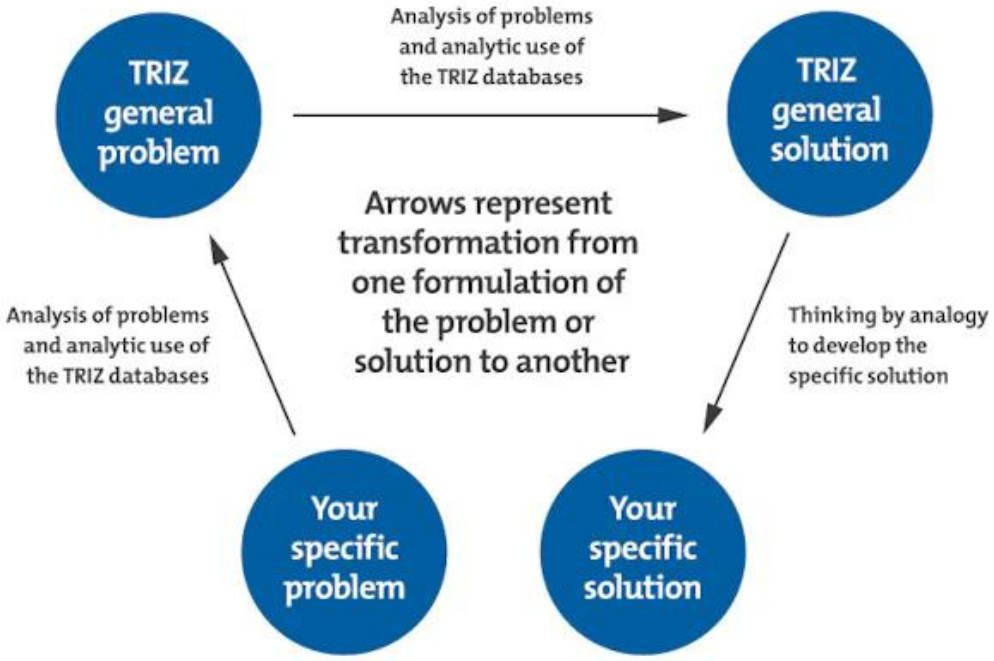
This reduces **time available for patient interaction**.

Problem statement:

How might we reduce the time doctors spend on documentation while maintaining or improving medical record quality?

https://www.triz-consulting.de/wp-content/uploads/2021/09/Contradiction_Matrix_V11.pdf

then https://triz.org/principles/



ANALOGICAL INSPIRATION

Problem

Another segment has solved the problem

Transfer of innovation

EXERCISE

Use errors sometimes occur because **device interfaces might be confusing**, especially in busy, noisy and stressfull hospital setups (e.g. emergencies).

Problem statement:

How might we reduce use errors caused by poorly designed user interfaces?

Think of another industry/segment where complex user interfaces are used and present significant safety risks for users and/or customers.

Identify 2 good practices from that industry and apply them to a medical device interface.

CONVERGE

Technique	Principle	When It Is Useful
Idea Clustering	Similar ideas are grouped into themes	Organizing large idea sets
Dot Voting	Participants vote for preferred ideas using dots or marks	Quick prioritization in groups
Impact–Feasibility Matrix	Ideas are evaluated based on impact and feasibility	Early-stage filtering
Concept Screening Matrix (Pugh Matrix)	Systematic comparison of ideas against criteria	Engineering decision making
Narrowing by Constraints	Ideas must satisfy key constraints (cost, time, resources)	Realistic project selection

IDEAS CLUSTERING

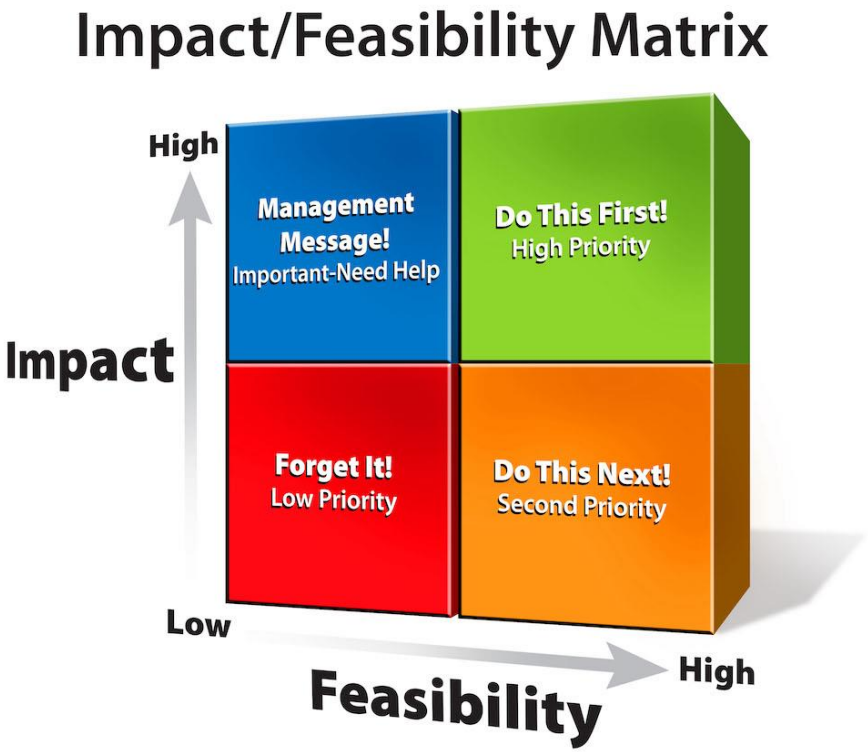
DOT VOTING RULES

- **One dot = one vote**
- **Limited number of dots** per person (e.g. 3–5)
- **Vote silently**
- **You may split dots** or stack them on one idea
- **Time-box the vote** (1–2 minutes)

After voting:

- **Top ideas move forward**
- **No debate until voting is over**

IMPACT vs FEASIBILITY



IMPACT VS EFFORT

CONCEPT SCREENING

SELECTION CRITERIA	CONCEPT VARIANTS							
	A	B	C	D	E	F	G	REF.
Ease of Handling	0	0	–	0	0	–	–	0
Ease of Use	0	–	–	0	0	+	0	0
Number Readability	0	0	+	0	+	0	+	0
Dose Metering	+	+	+	+	+	0	+	0
Load Handling	0	0	0	0	0	+	0	0
Manufacturing Ease	+	–	–	0	0	–	0	0
Portability	+	+	–	–	0	–	–	0
PLUSES	3	2	2	1	2	2	2	
SAMES	4	3	1	5	5	2	3	
MINUSES	0	2	4	1	0	3	2	
NET	3	0	–2	0	2	–1	0	
RANK	1	3	7	5	2	6	4	
CONTINUE?	Yes	Yes	No	No	Yes	No	Yes	

NARROWING BY CONSTRAINS

Questions

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